SGN – Assignment #2

MARTIN Mihnea Stefan, 223840

Disclaimer: The story plot contained in the following three exercises is entirely fictional.

Exercise 1: Uncertainty propagation

The Prototype Research Instruments and Space Mission Technology Advancement (PRISMA) is a technology in-orbit test-bed mission for demonstrating Formation Flying (FF) and rendezvous technologies, as well as flight testing of new sensors and actuator equipment. It was launched on June 15, 2010, and it involves two satellites: Mango (Satellite 1, ID 36599), the chaser, and Tango (Satellite 2, ID 36827), the target.

You have been provided with an estimate of the states of Satellites 1 and 2 at the separation epoch t_{sep} = `2010-08-12T05:27:39.114` (UTC) in terms of mean and covariance, as reported in Table 1. Assume Keplerian motion can be used to model the spacecraft dynamics.

1. Propagate the initial mean and covariance for both satellites within a time grid going from t_{sep} to $t_{sep} + N\,T_1$, with a step equal to T_1, where T_1 is the orbital period of satellite 1 and $N = 10$, using both a Linearized Approach (LinCov) and the Unscented Transform (UT). We suggest to use $\alpha = 0.1$ and $\beta = 2$ for tuning the UT in this case.

2. Considering that the two satellites are in close formation, you have to guarantee a sufficient accuracy about the knowledge of their state over time to monitor potential risky situations. For this reason, at each revolution, you shall compute:

   - the norm of the relative position (Δr), and
   - the sum of the two covariances associated to the position elements of the states of the two satellites (P_{sum})

   The critical conditions which triggers a collision warning is defined by the following relationship:
   $$\Delta r < 3\sqrt{\max(\lambda_i(P_{\text{sum}}))}$$
   where $\lambda_i(P_{\text{sum}})$ are the eigenvalues of P_{sum}. Identify the revolution N_c at which this condition occurs and elaborate on the results and the differences between the two approaches (UT and LinCov).

3. Perform the same uncertainty propagation process on the same time grid using a Monte Carlo (MC) simulation *. Compute the sample mean and sample covariance and compare them with the estimates obtained at Point 1). Provide the plots of:

   - the time evolution for all three approaches (MC, LinCov, and UT) of $3\sqrt{\max(\lambda_i(P_{r,\text{i}}))}$ and $3\sqrt{\max(\lambda_i(P_{v,\text{i}}))}$, where $i = 1, 2$ is the satellite number and P_r and P_v are the 3x3 position and velocity covariance submatrices.
   - the propagated samples of the MC simulation, together with the mean and covariance obtained with all methods, projected on the orbital plane.

   Compare the results and discuss on the validity of the linear and Guassian assumption for uncertainty propagation.

*Use at least 100 samples drawn from the initial covariance

Table 1: Estimate of Satellite 1 and Satellite 2 states at t_0 provided in ECI J2000.

Parameter	Value
Ref. epoch t_{sep} [UTC]	2010-08-12T05:27:39.114
Mean state $\hat{x}_{0,\text{sat1}}$ [km, km/s]	$\hat{r}_{0,\text{sat1}} =$ [4622.232026629, 5399.3369588058, -0.0212138165769957] $\hat{v}_{0,\text{sat1}} =$ [0.812221125483763, -0.721512914578826, 7.42665302729053]
Mean state $\hat{x}_{0,\text{sat2}}$ [km, km/s]	$\hat{r}_{0,\text{sat2}} =$ [4621.69343340281, 5399.26386352847, -3.09039248714313] $\hat{v}_{0,\text{sat2}} =$ [0.813960847513811, -0.719449862738607, 7.42706066911294]
Covariance P_0 [km^2, km^2/s, km^2/s^2]	$\begin{bmatrix} +5.6e-7 & +3.5e-7 & -7.1e-8 & 0 & 0 & 0 \\ +3.5e-7 & +9.7e-7 & +7.6e-8 & 0 & 0 & 0 \\ -7.1e-8 & +7.6e-8 & +8.1e-8 & 0 & 0 & 0 \\ 0 & 0 & 0 & +2.8e-11 & 0 & 0 \\ 0 & 0 & 0 & 0 & +2.7e-11 & 0 \\ 0 & 0 & 0 & 0 & 0 & +9.6e-12 \end{bmatrix}$

Part 1

1. **Linearized Approach** (LinCov) Once the problem has been initialized, in order to obtain the propagated mean and propagated covariance, the Keplerian propagator has been used, from which was also obtained the Jacobian of f at $\hat{\mathbf{x}}'$, as the propagated mean coincides with the mean of the final distribution, for the linear approximation.

$$\hat{\mathbf{x}} = \mathbf{f}(\hat{\mathbf{x}}') \tag{1}$$

   It was assumed at the beginning of the process that $\hat{\mathbf{x}}_{\mathbf{0}} = \mathbf{x_0}^*$, as $\hat{\mathbf{x}}$ must respect Eqs. 2

$$\begin{cases} \hat{\mathbf{x}}(t) = E[\mathbf{x}(t)] \\ \mathbf{x}(t) = \mathbf{x}^*(t) + \delta\mathbf{x}(t) \end{cases} \tag{2}$$

   To obtain the propagated covariance, the STM matrix is introduced in Eq. 3. In this equation $P(t_0)$ is the initial covariance P_0.

$$\mathbf{P}(t) = \mathbf{\Phi}(t_0, t)\mathbf{P}(t_0)\mathbf{\Phi}^T(t_0, t) \tag{3}$$

   This process has been used for both spacecraft and for each time step in the time grid, which was defined by the problem.

2. **Unscented Transform**(UT) For the case of Unscented Transform, the general UT algorithm has been used, for both spacecraft. The algorithm was implemented, similar to the LinCov case, at each time instance.

   I) The Sigma Points have been calculated through Eq. 4. In the equation, n has been chosen 6, as it is the size of matrix $\mathbf{P_0}$ and the scaling parameter was calculated as $\lambda = \alpha^2(n + k) - n$, where k was considered zero, and α was taken equal to 0.1, as suggested in the text. The index i represents the index of the column for both the sigma points matrix and for $(\sqrt{(n + \lambda)P_0})_i$.

$$\begin{cases} \mathbf{x}_0 = \hat{\mathbf{x}} \\ \mathbf{x_i} = \hat{\mathbf{x}} + (\sqrt{(n + \lambda)\mathbf{P_0}})_i & i = 1, ..., n \\ \mathbf{x_i} = \hat{\mathbf{x}} - (\sqrt{(n + \lambda)\mathbf{P_0}})_i & i = n + 1, ..., 2n \end{cases} \tag{4}$$

II) Once the sigma points have been found, the weights have been calculated by the formulas in Eq. 5, in which β was considered 2, as suggested. In Eq. 5 the superscripts *(m)* and *(c)* refer to the weight used for finding the mean, and the one used for finding the covariance.

$$\begin{cases} \mathbf{W_0}^{(m)} = \lambda/(n+\lambda) \\ \mathbf{W_0}^{(c)} = \lambda/(n+\lambda) + (1-\alpha^2+\beta) \\ \mathbf{W_i}^{(m)} = \mathbf{W_i}^{(c)} = \dfrac{1}{2(n+\lambda)} \qquad\qquad i = 1, ..., 2n \end{cases} \tag{5}$$

III) The weighted sample mean and covariance were calculated through Eq. 6 and Eq. 7, and the results were calculated by propagating the sigma points($\mathbf{y_i}$) through the Keplerian propagator.

$$\hat{\mathbf{y}} = \sum_{i=0}^{2n} \mathbf{W_i}^{(m)} \mathbf{y_i} \tag{6}$$

$$\mathbf{P_y} = \sum_{i=0}^{2n} \mathbf{W_i}^{(c)} (\mathbf{y_i} - \hat{\mathbf{y}})(\mathbf{y_i} - \hat{\mathbf{y}})^T \tag{7}$$

For the time grid indicated, the difference between UT and LinCov is of the order of 10^{-4}, but if it were to have a larger time grid(having a longer propagation), the difference would increase as well.

Part 2

In order to check the proximity between the two satellites, the criticality condition was used, by inserting the values already obtained in **Part 1** for each method. The critical condition (Eq. 8), was evaluated for each value of Δr and $\mathbf{P_{Sum}}$ until now. If the condition was violated, the corresponding orbit was recorded. Solutions were computed using both the LinCov and Unscented Transform (UT) methods, and the results were subsequently compared.

$$\Delta r < 3\sqrt{max(\lambda_i(\mathbf{P}_{sum}))} \tag{8}$$

In Eq. 8 Δr is the norm of the difference between the mean states of the two satellites. The critical condition (Eq. 8) was triggered in the same orbit (the 4^{th}) for both methods, occurring six times in each case. This indicates that, for the chosen initial values, the conditions are nearly identical for both methods. The error of relative position Δr between the two methods is of the order of magnitude of 10^{-6}, which would explain why the two methods offer similar results. However, as previously stated, if the propagation time increases, the behaviour of the two methods would increment.

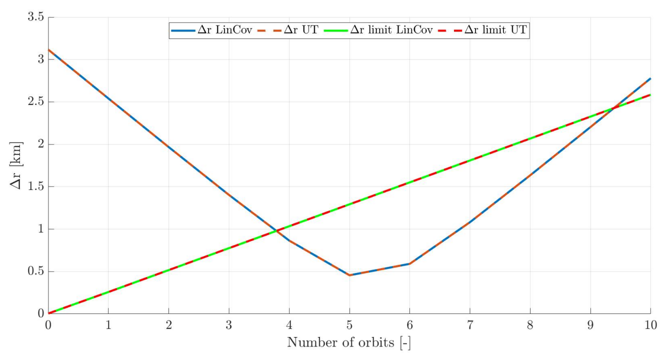

Figure 1: Δr for and Δr limit for LinCov and UT

Part 3

The Monte Carlo [MC] simulation is conducted using the *Matlab* functions *mvnrnd, cov* and *mean*. *mvnrnd* was used to generate the initial conditions, considering 750 samples. The result was propagated, and then the other two functions were used to calculate the mean and the covariance at each instance. The number of samples was considered to have a relatively high value in order to enhance the accuracy. If a lower level of accuracy is required or desired, a smaller number of samples could be chosen.
The derived solutions for the covariance and mean states, were converted into a local reference

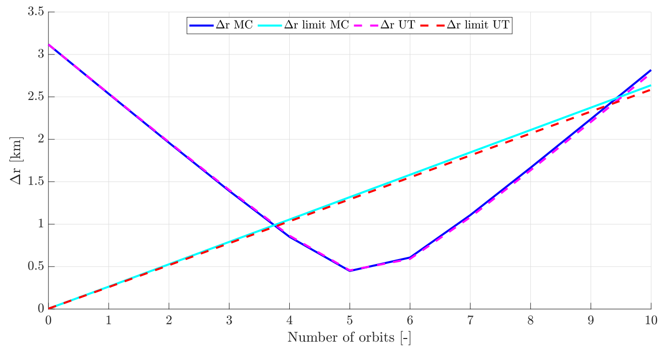

Figure 2: Δr and Δr limit for MC and UT

frame, defined for each satellite and the final results are shown in 4 and 5. From Fig. 2, a slight difference is observed between the UT method and MC method, regarding Δr and the limit evolution of the critical condition. This may be attributable to the degree of linearity of the problem, or the sample count taken for the Monte Carlo simulation. An improved solution can be derived also from propagating the MC with 1000 samples, however, it is computationally demanding and it necessitates a considerable amount of time.

The error may result from the initial values, as can be seen in Fig. 3. for Tango, the 3 methods produce similar results, while for Mango both the Position Covariance and the Velocity Covariance seem to slightly diverge for the Monte Carlo simulation. Thus, the error can originate from the creation of the random initial data, which is based on the initial states of the two satellites.

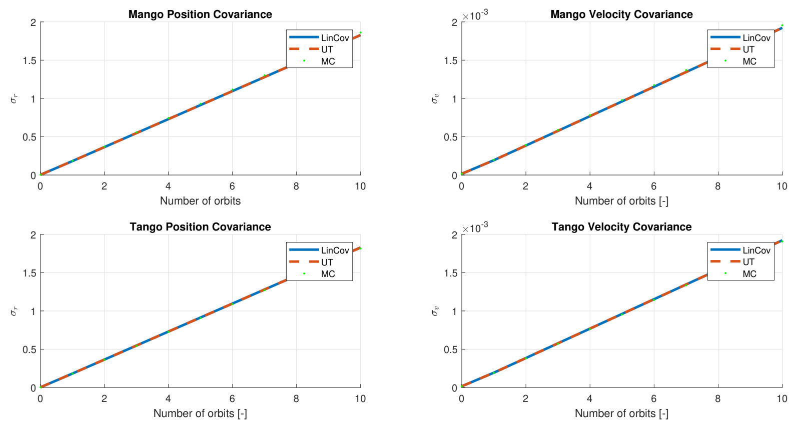

Figure 3: Position and Velocity covariance for Tango and Mango, for LinCov, UT and MC

The unscented transform exhibits superior performance with higher precision and it is not as computationally demanding as the Monte Carlo, obtaining a result with a higher precision. The LinCov results in this case are also favorable and the difference between it and the UT are small enough to consider as a viable solution. However, for more complex problems the final error would increase, thus for those types of problem, the only method that would end in an adequate solution is the UT.

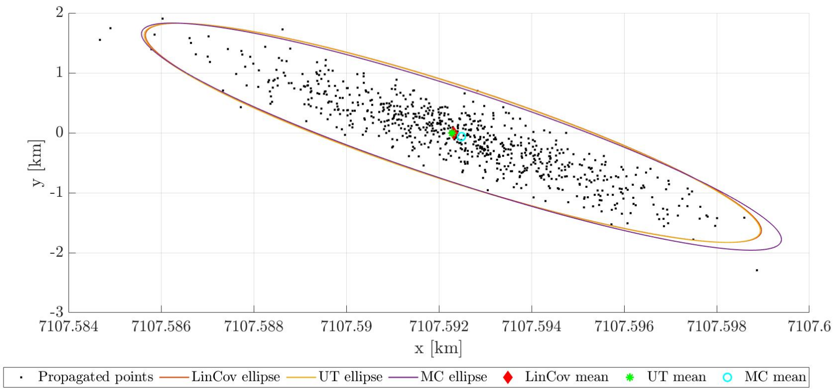

Figure 4: Mango mean points for each method, with their respective ellipses

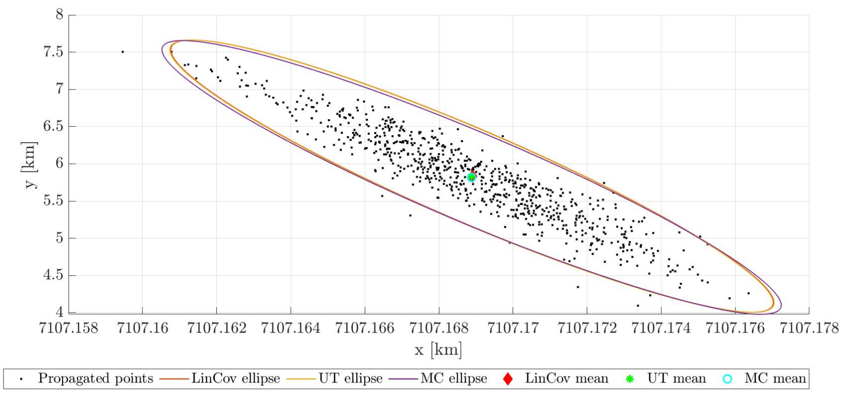

Figure 5: Tango mean points for each method, with their respective ellipses

Figures 5 and 4 display the mean states for each method and the covariance with the respective ellipses. The differences between the three methods are showcased, particularly in the case of Tango's mean state, which is shifted to the left in the Monte Carlo results, compared to the other two methods

Exercise 2: Batch filters

You have been asked to track Mango to improve the accuracy of its state estimate. To this aim, you shall schedule the observations from the two ground stations reported in Table 2.

1. *Compute visibility windows.* By using the mean state reported in Table 1 and by assuming Keplerian motion, predict the trajectory of the satellite over a uniform time grid (with a time step of 60 seconds) and compute all the visibility time windows from the available stations in the time interval from t_0 = `2010-08-12T05:30:00.000` (UTC) to t_f = `2010-08-12T11:00:00.000` (UTC). Plot the resulting predicted Azimuth and Elevation profiles in the visibility windows.

2. *Simulate measurements.* The Two-Line Elements (TLE) set of Mango are reported in Table 3 (and in WeBeep as 36599.3le). Use **SGP4** and the provided TLEs to simulate the measurements acquired by the sensor network in Table 2 by:

   I) Computing the spacecraft position over the visibility windows identified in Point 1 and deriving the associated expected measurements.

   II) Simulating the measurements by adding a random error to the expected measurements (assume a Gaussian model to generate the random error, with noise provided in Table 2). Discard any measurements (i.e., after applying the noise) that does not fulfill the visibility condition for the considered station.

3. *Solve the navigation problem.* Using the measurements simulated at the previous point:

   I) Find the least squares (minimum variance) solution to the navigation problem without a priori information using
      - the epoch t_0 as reference epoch;
      - the reference state as the state derived from the TLE set in Table 3 at the reference epoch;
      - the simulated measurements calculated for the KOUROU ground station only;
      - pure Keplerian motion to model the spacecraft dynamics.

   II) Repeat step 3a by using all simulated measurements from both ground stations.

   III) Repeat step 3b by using J2-perturbed motion to model the spacecraft dynamics.

4. Provide the obtained navigation solutions and elaborate on the results, comparing the different solutions.

5. Select the best combination of dynamical model and ground stations and perform the orbit determination for the other satellite.

Part 1

The Visibility time windows are determined based on the elevation angle of the satellite seen from the ground station. This angle must respect the minimum condition of minimum elevation angle, below which the satellite is not visible due to possible obstructions or curvature of the Earth.

The computation of the visibility time windows for the Mango satellite is based on its position calculated from each ground station, Kourou and Svalbard. The initial position vectors have been converted into Earth-Centered Inertial frame, and from there into Topocentric reference frame in order to obtain the angular measurements. After the Topocentric coordinates have been obtained, the $cspice\_xfmsta$ has been used, which offers as results the satellite's

Table 2: Sensor network to track Mango and Tango: list of stations, including their features.

Station name	KOUROU	SVALBARD
Coordinates	LAT = 5.25144° LON = -52.80466° ALT = -14.67 m	LAT = 78.229772° LON = 15.407786° ALT = 458 m
Type	Radar (monostatic)	Radar (monostatic)
Provided measurements	Az, El [deg] Range (one-way) [km]	Az, El [deg] Range (one-way) [km]
Measurements noise (diagonal noise matrix R)	$\sigma_{Az,El}$ = 100 mdeg σ_{range} = 0.01 km	$\sigma_{Az,El}$ = 125 mdeg σ_{range} = 0.01 km
Minimum elevation	10 deg	5 deg

Table 3: TLE of Mango.

```
1␣36599U␣10028B␣␣␣10224.22752732␣-.00000576␣␣00000-0␣-16475-3␣0␣␣9998
2␣36599␣098.2803␣049.5758␣0043871␣021.7908␣338.5082␣14.40871350␣␣8293
```

Table 4: TLE of Tango.

```
1␣36827U␣10028F␣␣␣10224.22753605␣␣.00278492␣␣00000-0␣␣82287-1␣0␣␣9996
2␣36827␣098.2797␣049.5751␣0044602␣022.4408␣337.8871␣14.40890217␣␣␣␣55
```

coordinates in Range, Azimuth and Elevation. However, azimuth is bounded by the interval $[0, 360]$, thus the values obtained must be converted into degrees, and adapted to belong to the interval. The visibility time windows are based on the ground station capabilities, specifically the minimum elevation that each station can perceive. Thus, the visibility time windows are defined by the points in the time grid that satisfy $El_{Mango} \geqslant El_{Station_{min}}$.

The results obtained for each station are shown in Fig. 6 and 7 , and they represent the Azimuth and Elevation registered from each station in the visibility window.

Part 2

To determine the satellite's position over the visibility window, TLEs are utilized in order to obtain the reference epoch of the satellite, which was used to define the new time grid with respect to the data obtained from the TLEs. Additionally, in order to use the transformation from the TEME frame to the ECI frame, a correction nutation was needed, which was taken from [1]. The final values were computed into radians.

The **SGP4** function is used to obtain the state vector of the satellite in the TEME reference frame, which uses as input the time grid defined earlier, and the data retrieved from the TLEs. Subsequently, the states in TEME are converted into the ECI frame.

The required angular coordinates necessary for simulations, are derived from a process similar to the one described in **Part 1**, with the introduction of random error, generated through a Gaussian model distribution, following the formula in Eq. 9

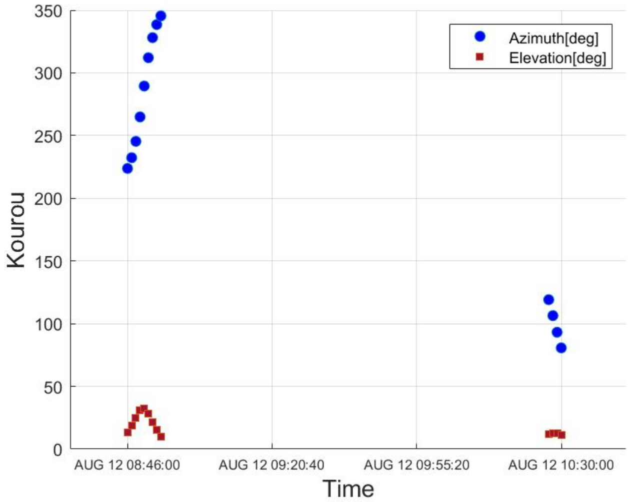

Figure 6: Mango Azimuth and Elevation from Kourou

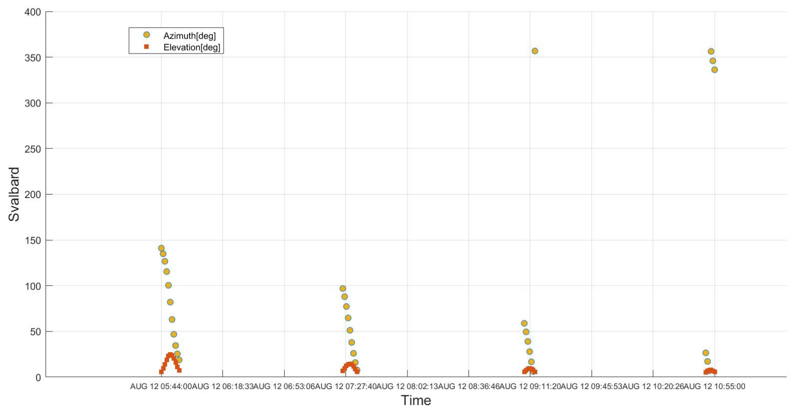

Figure 7: Mango Azimuth and Elevation from Svalbard

$$\mathbf{R} = \begin{bmatrix} \sigma_\rho^2 & 0 & 0 \\ 0 & \sigma_{Az}^2 & 0 \\ 0 & 0 & \sigma_{El}^2 \end{bmatrix} \tag{9}$$

The modeled measured angular state is then compared with the minimum elevation capability of the station, and the visibility time is given by the instances of the visibility time obtained in **Part 1**, for which the measurement respects the elevation constraint.

Part 3

1. The least square solution for the value of x_k that minimizes the sum of the square of the residuals must be found. Firstly, the Weighted method was used, the weight matrix being obtained as in formula 10. Now the objective is to find the $\delta\tilde{\mathbf{x}}_\mathbf{k}$ that minimizes J, whose

formula is written in Eq. 11

$$\mathbf{W} = \begin{bmatrix} \frac{1}{\sigma_{range}^2} & 0 & 0 \\ 0 & \frac{1}{\sigma_{Az}^2} & 0 \\ 0 & 0 & \frac{1}{\sigma_{EL}^2} \end{bmatrix} \tag{10}$$

In Eq. 11 $\delta\mathbf{y}$ is the variation of the angular measurements obtained beforehand, with respect to the measurements obtained from the reference trajectory (x_k^*,y_i^*) and H_i is the partial derivative of the measurements with the states on the initial trajectory.

$$\mathbf{J} = \frac{1}{2}(\delta\mathbf{y} - \mathbf{H}\delta\mathbf{x}_k)^T\mathbf{W}(\delta\mathbf{y} - \mathbf{H}\delta\mathbf{x_k}) \tag{11}$$

$$\begin{pmatrix} \delta\mathbf{y_1} \\ \delta\mathbf{y_2} \\ \vdots \\ \delta\mathbf{y_l} \end{pmatrix} = \begin{bmatrix} \tilde{H}_1\Phi(t_k, t_1) \\ \tilde{H}_2\Phi(t_k, t_2) \\ \vdots \\ \tilde{H}_l\Phi(t_k, t_l) \end{bmatrix} \delta x_k + \begin{pmatrix} \epsilon_\mathbf{1} \\ \epsilon_\mathbf{2} \\ \vdots \\ \epsilon_\mathbf{l} \end{pmatrix} \tag{12}$$

which is equivalent to

$$\delta\mathbf{y} = \mathbf{H}\delta\mathbf{x_k} + \epsilon \tag{13}$$

This solution is obtained using the *Matlab* function *lsqnonlin* and it is obtainable from Eq. 14, representing the minimum variance solution. The state is propagated at each time step, and the resulting solution is converted into angular measurements, from which the initial measurements from the Kourou station are subtracted.

$$\delta\hat{\mathbf{x}}_\mathbf{k} = (\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{W}\delta\mathbf{y} \tag{14}$$

If desired, the covariance can be estimated in a similar manner, by using the Eq. 15

$$\mathbf{P_k} = (\mathbf{H^T}\mathbf{R}^{-1}\mathbf{H})^{-1} \tag{15}$$

2. The process for both ground stations, and the results are shown in Table 5, the difference is that now the measurements of both stations are taken into account, and thus the results are more exact.

3. In addition to the previous step, to the Keplerian propagator was added the J_2 perturbation, while the rest of the process has been similar. The results of this method are shown in Table 5.

Part 4

From Table 5 and Table 6, it is evident that the optimal solution involves measuring from both stations with the inclusion of the J_2 perturbation. This improvement was anticipated because the stations are geographically separated and possess different noise characteristics. Additionally, the radars lack the capability to discern the direction from which the signals arrive, so utilizing two distinct stations helps to minimize this error.

Another potential source of error arises from the Weight Matrix (Equation 10), which is computed based on the noise matrix of each ground station. Consequently, the significance of each angular measurement varies between stations.

From Table 6, the impact of accounting for the J_2 perturbation is significant. This discrepancy can be attributed to the more accurate predictions provided by the J_2 Keplerian

Kourou	Kourou and Svalbard	J_2 perturbation	Units
4662.87	4666.33	4685.42	[km]
5255.46	5248.93	5238.58	[km]
1010.50	1039.68	1042.57	[km]
0.0903	0.09205	0.08156	[km/s]
-1.4969	-1.54787	-1.55659	[km/s]
7.3619	7.35020	7.34493	[km/s]

Table 5: Least Square Solutions with measurements from Kourou, Both Station and including J2 perturbation

	Koruou	Kourou and Svalbars	J_2 perturbation	Units
Δr	42.52617	21.80373	0.16735	[km]
Δv	0.06259	0.01457	$2.357 \cdot 10^{-4}$	[km/s]

Table 6: Error of each method

propagator, which accounts for the Earth's oblateness and its effects on satellite orbits. The error is expected to increase further over extended propagation periods.

Part 5

The results from **Part 4** highlight the efficacy of using two ground stations and including the J_2 perturbation, for obtaining an orbit determination. This approach remains effective even when the Svalbard station is considered alone with the J_2 perturbation.
In Table 7 the states of the Tango measurements are shown, while in Table 8 the errors are shown, reaffirming the importance of including two ground stations and accounting for the perturbation in the propagating model.

Tango States	Units
4684.92	[km]
5238.76	[km]
1038.74	[km]
0.083819	[km/s]
-1.554155	[km/s]
7.345680	[km/s]

Table 7: States of Tango measurements

	J_2 perturbation	Units
Δr	0.722949	[km]
Δv	$6.2729 \cdot 10^{-4}$	[km/s]

Table 8: Error of Tango measurements

Exercise 3: Sequential filters

According to the Formation Flying In Orbit Ranging Demonstration experiment (FFIORD), PRISMA's primary objectives include testing and validation of GNC hardware, software, and algorithms for autonomous formation flying, proximity operations, and final approach and recede operations. The cornerstone of FFIORD is a Formation Flying Radio Frequency (FFRF) metrology subsystem designed for future outer space formation flying missions.

FFRF subsystem is in charge of the relative positioning of 2 to 4 satellites flying in formation. Each spacecraft produces relative position, velocity and line-of-sight (LOS) of all its companions.

You have been asked to track Mango to improve the accuracy of the estimate of its absolute state and then, according to the objectives of the PRISMA mission, validate the autonomous formation flying navigation operations by estimating the relative state between Mango and Tango by exploiting the relative measurements acquired by the FFRF subsystem. The Two-Line Elements (TLE) set of Mango and Tango satellites are reported in Tables 3 and 4 (and in WeBeep as 36599.3le, and 36827.3le).

The relative motion between the two satellites can be modelled through the linear, Clohessy-Wiltshire (CW) equations†

$$\begin{aligned} \ddot{x} &= 3n^2x + 2n\dot{y} \\ \ddot{y} &= -2n\dot{x} \\ \ddot{z} &= -n^2z \end{aligned} \tag{16}$$

where x, y,and z are the relative position components expressed in the LVLH frame, whereas n is the mean motion of Mango, which is assumed to be constant and equal to:

$$n = \sqrt{\frac{GM}{R^3}} \tag{17}$$

where R is the position of Mango at t_0.

The unit vectors of the LVLH reference frame are defined as follows:

$$\hat{i} = \frac{r}{r}, \quad \hat{j} = \hat{k} \times \hat{i}, \quad \hat{k} = \frac{h}{h} = \frac{r \times v}{\|r \times v\|} \tag{18}$$

To perform the requested tasks you should:

1. *Estimate Mango absolute state.* You are asked to develop a sequential filter to narrow down the uncertainty on the knowledge of Mango absolute state vector. To this aim, you shall schedule the observations from the SVALBARD ground station‡ reported in Table 2, and then proceed with the state estimation procedure by following these steps:

   I) By using the mean state reported in Table 1 and by assuming Keplerian motion, predict the trajectory of the satellite over a uniform time grid (with a time step of 5 seconds) and compute the first visibility time window from the SVALBARD station in the time interval from t_0 = `2010-08-12T05:30:00.000` (UTC) to t_f = `2010-08-12T06:30:00.000` (UTC).

   II) Use **SGP4** and the provided TLE to simulate the measurements acquired by the SVALBARD station for the Mango satellite only. For doing it, compute the spacecraft position over the visibility window using a time-step of 5 seconds, and derive the associated expected measurements. Finally, simulate the measurements by adding a random error (assume a Gaussian model to generate the random error, with noise provided in Table 2).

†Notice that the system is linear, therefore it has an analytic solution of the state transition matrix Φ

‡Note that these are the same ones computed in Exercise 2

III) Using an Unscented Kalman Filter (UKF), provide an estimate of the spacecraft state (in terms of mean and covariance) by sequentially processing the acquired measurements in chronological order. Plot the time evolution of the error estimate together with the 3σ of the estimated covariance for both position and velocity.

2. *Estimate the relative state.* To validate the formation flying operations, you are also asked to develop a sequential filter to narrow down the uncertainty on the knowledge of the relative state vector. To this aim, you can exploit the relative azimuth, elevation, and range measurements obtained by the FFRF subsystem, whose features are reported in Table 9, and then proceed with the state estimation procedure by following these steps:

   I) Use **SGP4** and the provided TLEs to propagate the states of both satellites at epoch t_0 in order to compute the relative state in LVLH frame at that specific epoch.

   II) Use the relative state as initial condition to integrate the CW equations over the time grid defined in Point 1a. Finally, simulate the relative measurements acquired by the Mango satellite through its FFRF subsystem by adding a random error to the expected measurements. Assume a Gaussian model to generate the random error, with noise provided in Table 9.

   III) Consider a time interval of 20 minutes starting from the first epoch after the visibility window (always with a time step of 5 seconds). Use an UKF to provide an estimate of the spacecraft relative state in the LVLH reference frame (in terms of mean and covariance) by sequentially processing the measurements acquired during those time instants in chronological order. Plot the time evolution of the error estimate together with the 3σ of the estimated covariance for both relative position and velocity.

3. *Reconstruct Tango absolute covariance.* Starting from the knowledge of the estimated covariance of the absolute state of Mango, computed in Point 1, and the estimated covariance of the relative state in the LVLH frame, you are asked to provide an estimate of the covariance of the absolute state of Tango. You can perform this operation as follows:

   I) Pick the estimated covariance of the absolute state of Mango at the last epoch of the visibility window, and propagate it within the time grid defined in Point 2c.

   II) Rotate the estimated covariance of the relative state from the LVLH reference frame to the ECI one within the same time grid.

   III) Sum the two to obtain an estimate of the covariance of the absolute state of Tango. Plot the time evolution of the 3σ for both position and velocity and elaborate on the results.

Table 9: Parameters of FFRF.

Parameter	Value
Measurements noise (diagonal noise matrix R)	$\sigma_{Az,El} = 1$ deg $\sigma_{range} = 1$ cm

Part 1

1. The propagation and visibility time window were computed as described in **Exercise 2**. Initially, the ECI coordinates of the satellite were computed at each time instance. Subsequently, these coordinates were transformed relative to the station's ECI coordinates to derive the angular measurements. The visibility window corresponds to the time instances during which the satellite's elevation angle exceeds the station's minimum elevation threshold.

2. The measurements acquired by the Svalbard station were simulated by following the process used in **Exercise 2**. The satellite's state in the TEME frame was obtained using the **SGP4** function and then transformed into the ECI frame, with an added nutation correction. The resulting data were converted into angular measurements and verified against the visibility time window. A random error was incorporated according to the matrix in Equation 9.

3. The process of implementing the Unscented Kalman Filter was similar to the implementation of the Unscented Transform in **Exercise 1**. Starting from the initial state of Mango, from which the sigma points were calculated by using Eq. 4, and the weights were obtained from Eq. 5. The sigma points were propagated using a Keplerian propagator, that included also the J_2 perturbation.
   Once the Unscented Transform was completed, the results were corrected using the filter. The measurements were simulated by propagating the sigma points. The measurements have to be updated for each time step, and for both y and x. This process was done through Eqs. 19 and 20

$$\mathbf{P_{y_k y_k}} = \sum_{i=0}^{2L} W_i^{(c)} [\mathbf{y_{i,k|k-1}} - \mathbf{\hat{y_k}}^-][\mathbf{y_{i,k|k-1}} - \mathbf{\hat{y_k}}^-]^T \tag{19}$$

$$\mathbf{P_{x_k y_k}} = \sum_{i=0}^{2L} W_i^{(c)} [\mathbf{x_{i,k|k-1}} - \mathbf{\hat{x_k}}^-][\mathbf{y_{i,k|k-1}} - \mathbf{\hat{y_k}}^-]^T \tag{20}$$

   Where $y_{i,k}$ are the sigma points in measurement space, $\mathbf{\hat{y_k}}$, is the mean of the measurements $y_{i,k}$, with $P_{y_k y_k}$ is its covariance, L is the dimension of augmented state and $P_{x_k y_k}$ is the cross covariance of the measurements with the states. If the matrix $\mathbf{P_{y_k y_k}}$ can be inverted, then the Kalman gain (Eq. 21) is calculated to correct the measurements, which is done following the Eqs. 22 and 23.

$$\mathbf{K} = \mathbf{P_{x_k y_k}} \mathbf{P_{y_k y_k}}^{-1} \tag{21}$$

   However, if the matrix $\mathbf{P_{y_k y_k}}$ is not invertible, the Kalman gain cannot be calculated, and thus the measurements are not updated and the process continues with the next iteration, keeping as output states from the filter the ones obtained through the Unscented Transform.

$$\mathbf{\hat{x_k}} = \mathbf{\hat{x_k}}^- + \mathbf{K}(\mathbf{y_k} - \mathbf{\hat{y_k}}^-) \tag{22}$$

$$\mathbf{P_k} = \mathbf{P_k^-} - K\mathbf{P_{y_k y_k}} \mathbf{K}^T \tag{23}$$

   The results were plotted as requested, in Fig. 8. It is observable that the error exhibits decreasing pattern. The error of order 10^{-1} for position and 10^{-4} for velocity are acceptable for nonlinear problems, and its convergence pattern is set from the beginning, as it tends to have a convergent behaviour.

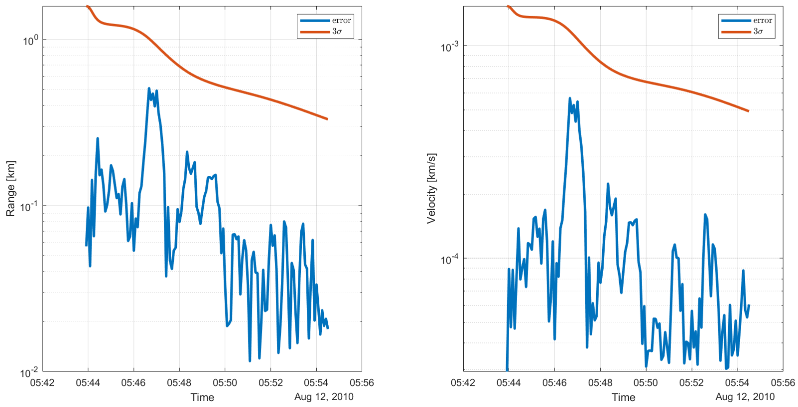

Figure 8: 3σ of the estimated covariance and estimated error(@J2000)

Part 2

1. Similarly to **Exercise 2**, the measurements have been computed for both satellites, obtaining their states in the ECI frame. This was obtained from the *measurements* function, with the use of **SGP4**. The initial measurements were taken from the provided TLEs of both satellites.

2. From their measured states, the relative states between the two satellites have been computed (Eq. 25) and converted into Local Vertical Local Horizon frame through the formulas in Eq. 24.

$$\begin{cases} \mathbf{i} = \dfrac{\mathbf{r_{ECI}}}{|\mathbf{r_{ECI}}|} \\ \mathbf{j} = |\mathbf{k} \times \mathbf{i}| \\ \mathbf{k} = \dfrac{\mathbf{r_{ECI}} \times \mathbf{v_{ECI}}}{|\mathbf{r_{ECI}} \times \mathbf{v_{ECI}}|} \end{cases} \tag{24}$$

$$\mathbf{x}_{Tango-Mango} = \mathbf{R}_{LVLH} \begin{pmatrix} \mathbf{r}_{ECI-Tango} - \mathbf{r}_{ECI-Mango} \\ \mathbf{v}_{ECI-Tango} - \mathbf{v}_{ECI-Mango} \end{pmatrix} \tag{25}$$

Where $\mathbf{R_{LVLH}}$ is the transformation matrix from ECI to LVLH, formed by the vectors in equation 24. The relative state obtained was introduced into an *ODE* propagator, which was defined using the Clohessy-Wiltshire equations, already given These equations are used to describe the relative state of a satellite with respect to another satellite(reference). The obtained results were then transformed into angular measurements(range,azimuth and elevation), to which a Gaussian model was used to generate the random error based on the noise provided. The noise matrix was firstly converted into the desired units, before being introduced into the calculations .

3. The new time interval has been defined, and the Tango's states and measurements have been taken with respect to it. The values are taken as the initial values for the UKF, and states and the covariance are propagated. Differently from the previous exercise, now also inside the UKF function, the sigma points have been propagated through the mean of **CW** propagator. The CW represents the relative state of a satellite with respect to the target satellite. Thus the nature of uncertainties could differ from the initial state, and thus a new initial value for the covariance matrix should be considered. The new matrix is diagonal and has the form given in Eq. 26:

$$\mathbf{P}_{0-CW} = \begin{pmatrix} 0.01 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.01 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.0001 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.0001 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.001 \end{pmatrix} \tag{26}$$

The initial states introduced into the UKF filter were considered to be the states obtained from the Clohessy-Wiltshire propagator at the beginning of the new time grid.

With the results obtained from the propagation, the errors and the 3σ have been calculated with respect to the initial vector of states obtained from the Clohessy - Wiltshire propagator for the entire time grid. The obtained results are shown in Fig. 9.

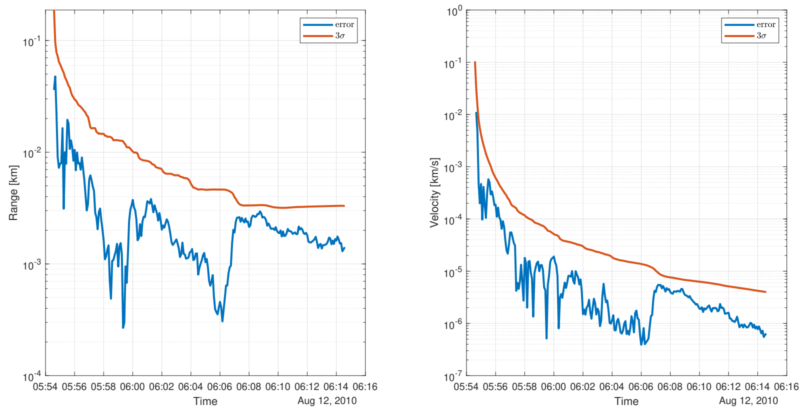

Figure 9: Time evolution of the error estimate and 3σ of the estimated covariance for the relative position and velocity(@LVLH J2000)

Part 3

A new time grid was defined with respect to the previous time grid, adding a new time step from $time - grid(1) - 5$ to $time - grid(1)$. For the new time grid, the mean states and mean covariances were propagated through the Unscented Kalman Filter and the Clohessy-Wiltshire propagator. The initial states of Tango for the propagation was considered to be the last states and covariance obtained from the UKF for the Mango satellite in **Part 1**.

The propagation results obtained from the Clohessy-Wiltshire were converted into ECI reference frame from LVLH, by using the inverse transformation of $\mathbf{R}$ (24). In this case, $\mathbf{R}$ was defined with respect to the states obtained from the aforementioned propagates states. The inverse transformation into ECI follows the Eq. (27) and is applied to the covariance matrix obtained in **Part 2**, resulting from the UKF propagation with the Clohessy-Wiltshire equations.

$$\mathbf{P}_{ECI} = \mathbf{R}^T \mathbf{P}_{CW} \mathbf{R} \tag{27}$$

The absolute covariance matrix required is $\mathbf{P}_{TAN-Absolute} = \mathbf{P}_{CW-Rotated} + \mathbf{P}_{Tango-Absolute}$, where $\mathbf{P}_{CW-Rotated}$ is the covariance obtained from CW propagation rotated in ECI frame and $\mathbf{P}_{Tango-Absolute}$ is the covariance propagated for the update time grid. The results of the evolution of the 3σ with time are shown in Fig. 10.

Fig. 10 shows a relatively steep increase in the value of 3σ with respect to time for the range measurements and a steep initial decrease for the velocity 3σ evolution.

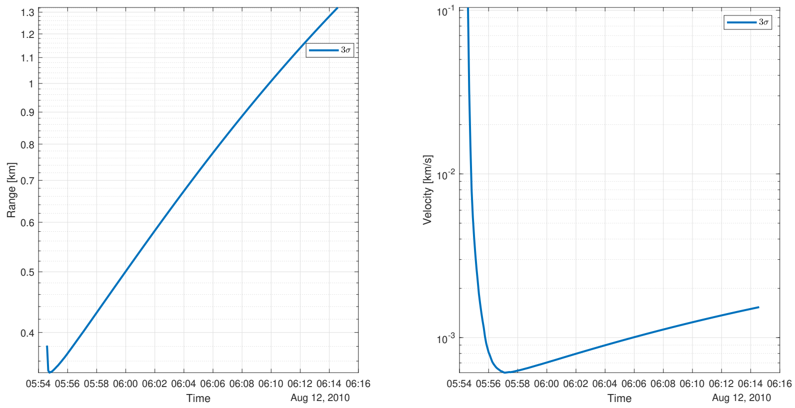

Figure 10: 3σ evolution with time for Tango's absolute position and velocity(J2000)

The increase in the range 3σ is represented by the accumulation of errors during the propagation process, as in this process the correction was not applied to the a priori mean state and covariance, and the values were just propagated through the Keplerian propagator. The values could be considered acceptable, however the efficiency of the Unscented Kalman Filter is proven by the difference in results, between the corrected values and the uncorrected values. This difference could prove to be paramount in flyby scenarios or tandem flying.

In the beginning the behaviour of the 3σ for velocity is similar to the corrected value, having a rapid decrease. However, the uncorrected value tends to increase in value after this, while the corrected value begin a convergent behaviour. The same order of magnitude as in the case of range is present.

References

[1] CelesTrak. *GPS Constellation Information*. Accessed: 2024-05-20. 2024 (cit. on p. 7).